

Lab 07

Design Pattern Lab

CSE 464



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1 Lab Exercise: Iterator Design Pattern

Objective: The objective of this lab is to understand and implement the **Iterator Design Pattern** by designing a collection that can be traversed without exposing its internal data structure.

Scenario

A university maintains a collection of students enrolled in a course. The students are stored internally by a `StudentCollection` class. The collection may use different internal data structures, such as an array or `ArrayList`, and the client should not need to know how the students are stored.

The system should allow users to traverse the students one by one using a standard interface. The client should be able to check whether another student is available and retrieve the next student without directly accessing the underlying collection or its indexes.

The design should also allow the internal representation of the collection to be changed in the future without requiring changes to the client code.

Tasks

- a) Implement the system in Java using the **Iterator Design Pattern**.
- b) Create an appropriate iterator interface containing operations for:
 - Checking whether another student is available.
 - Retrieving the next student.
- c) Implement a concrete iterator for the `StudentCollection` class.
- d) Ensure that the client does not directly access the internal data structure of `StudentCollection`.
- e) Add at least five students to the collection and traverse all students using the iterator.
- f) Demonstrate that the client can traverse the collection without knowing whether the students are stored internally using an array or an `ArrayList`.

Expected Classes

- `Student`
- `Iterator <T > (Interface)`
- `StudentIterator`
- `StudentCollection (Aggregate)`
- `Main`

Expected Structure

The design should follow the general structure:

```
<<interface>>
    Iterator
+ hasNext()
+ next()
    ^
    |
StudentIterator
    |
    v
StudentCollection
    + createIterator()
    |
    v
    Client
```

Additional Requirement

After completing the basic implementation, modify the internal storage of **StudentCollection** from an array to an **ArrayList** (or vice versa). The client code should continue to work without modification.

Note: Do not use Java's built-in `java.util.Iterator` for the main implementation. Create your own iterator interface to demonstrate the design pattern.

Submission Instructions

- Put all `.java` files inside a single folder.
- Compress the folder into a ZIP file.
- The ZIP file must be named using the following format:

`ID_Lab01.zip`